So $5\frac{1}{2}-3\frac{1}{4}=(5-3)+\left(\frac{1}{2}-\frac{1}{4}\right)$

$= 2+\left(\frac{2}{4}-\frac{1}{4}\right)$ (Since the equivalent fraction of $\frac{1}{2}$ is $\frac{2}{4}$)

$= 2+\frac{1}{4}=2\frac{1}{4}$ litres

Example 15 Simplify: $9\frac{1}{4}-3\frac{5}{6}$

Solution Here 9 > 3 and $\frac{1}{4}<\frac{5}{6}$, So we proceed as follows:

We convert the mixed fraction into improper fraction and then subtract.

$9\frac{1}{4}=\frac{(9\times4)+1}{4}=\frac{37}{4}$

and $3\frac{5}{6}=\frac{(3\times6)+5}{6}=\frac{23}{6}$

Common multiple of 4 and 6 is 12.

Now, $\frac{37}{4}-\frac{23}{6}=\frac{37\times3}{12}-\frac{23\times2}{12}$

$=\frac{111}{12}-\frac{46}{12}=\frac{65}{12}=5\frac{5}{12}$

Try these

i) Find the sum of $5\frac{4}{9}$ and $3\frac{1}{6}$

ii) Subtract $7\frac{1}{6}$ from $12\frac{3}{8}$.

iii) Subtract the sum of $6\frac{1}{6}$ and $3\frac{1}{5}$ from the sum of $9\frac{2}{3}$ and $2\frac{1}{2}$.

1.6 Multiplication of Fractions

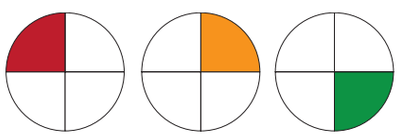
Think about the situation 1 (Multiplication of a fraction by a whole number)

Sunitha wanted to give $\frac{1}{4}$ kg of sweets to each of her 3 friends. So she went to a sweet stall and she asked the salesman to give three $\frac{1}{4}$ kg packets of sweets, how much sweet did she buy?

Solution Weight of three $\frac{1}{4}$ kg packets of sweets $=\frac{1}{4}+\frac{1}{4}+\frac{1}{4}=\frac{1+1+1}{4}=\frac{3}{4}$ kg



We can illustrate this in the following diagram. In each of these the shaded part is $\frac{1}{4}$ of a circle. Can you find the shaded parts in 3 circles together?



To know the fraction of shaded part in all the three circles, we add the fractions which are represented in each of the circles. So $3\times\frac{1}{4}=\frac{3}{4}$.

The adjacent circle represents $\frac{3}{4}$ parts of the circle.



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